ANNEX 1

CHAINSILIENCE AI

Glossary and Technical Implementation Annex

A governed agentic, knowledge-graph, and digital-twin framework for supply-chain disruption intelligence

Submitted as part of the project white paper for the International Data Science Olympiad (IDSOL) 2026

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Aligned with Whitepaper

Purpose of this annex
This annex defines the white paper's keywords, expands the proposed system workflow, and records the data, analytical, governance, and pilot details needed to interpret the architecture. It does not convert illustrative values or pilot targets into achieved results.

1. Annex Scope and Reading Guide

The main white paper presents the competition narrative and six-page architecture. This annex provides the supporting definitions and implementation detail. Its terminology and status labels follow the most recent revised white paper.

1.1 Status labels used throughout

Label	Meaning in this annex
Implemented / prototyped	A module or interface demonstrated by the current functional prototype, limited to what actually exists.
Planned	An architecture component, enterprise integration, simulation, scoring method, or controlled action proposed for later implementation.
Assumed	Required enterprise data, permissions, constraints, or scenario inputs supplied for design and evaluation.
Illustrative	A case-study value used to demonstrate workflow behavior; it is not a measured result or a claim about a real company.

1.2 Workflow at a glance

1. Detect and verify	2. Map exposure	3. Analyze impact	4. Plan and approve
Collect permitted sources; filter relevance; extract and cross-check claims.	Resolve entities; traverse Supplier -> Part -> Product -> Order dependencies.	Update the digital twin; calculate inventory, delays, cost, and scenario risk.	Retrieve approved options; test feasibility; compare mitigations; obtain authorization.

1.3 Core design boundary

Hybrid decision-support architecture

LLMs interpret language, retrieve and organize evidence, propose candidate actions, and draft explanations. Databases preserve records; the graph stores relationships; deterministic services calculate quantities; simulation estimates scenario outcomes; optimization compares feasible actions; policy controls permissions; and named humans retain decision authority.

2. Concise Keyword Dictionary

The definitions below correspond exactly to the ten keywords listed near the start of the revised white paper. Each entry states what the term means and why it matters in Chainsilience AI.

- **Supply-chain resilience:** The ability of a supply network to anticipate, absorb, adapt to, and recover from disruption while maintaining critical service.
  - **Use and significance:** It is the system's intended outcome. Chainsilience AI improves resilience by shortening detection and assessment time, quantifying exposure, and comparing feasible mitigations before operations stop.
- **Agentic AI:** AI organized into goal-directed agents that reason over evidence and use permitted tools to complete bounded tasks.
  - **Use and significance: A non-AI Scout agent polls approved RSS news feeds; the Extractor, Verifier, Mapper, Analyst, Planner, and Communicator are AI agents that divide the remaining workflow into auditable duties. Narrow permissions and human approval reduce the risk of unrestricted autonomous action.**
- **Digital twin:** A time-indexed computational representation of the current supply-chain network and its operating state.
  - **Use and significance:** The twin combines inventory, demand, orders, capacity, yield, lead times, routes, and recovery assumptions. It lets the system replay disruptions and compare no-action and mitigated scenarios on a reproducible snapshot.
- **Knowledge graph:** A graph of supply-chain entities and the relationships between them, with evidence, confidence, and temporal validity.

    - **Use and significance:** It connects an external event to affected suppliers, sites, parts, products, routes, production lines, and customer orders. This converts a general news alert into a company-specific exposure path.
- **Semiconductor risk:** Operational risk arising from concentrated fabrication and assembly capacity, specialized materials and equipment, long qualification cycles, geopolitical controls, and limited substitutability.
    - **Use and significance:** Chainsilience AI represents these dependencies and constraints so that alternatives are not recommended merely because they appear in public information.
- **Retrieval-augmented generation (RAG):** A method that retrieves relevant, current records and supplies them to an LLM before it generates an output.
    - **Use and significance:** RAG grounds event summaries, exposure explanations, and mitigation proposals in source-linked news and authorized enterprise records. During planning it retrieves approved suppliers, qualified substitutes, contracts, policies, capacity, routes, and previous responses.
- **Large language model (LLM) reasoning:** The use of an LLM to interpret unstructured language, relate retrieved evidence, propose options, and explain conclusions.
    - **Use and significance:** The LLM filters relevance, extracts events, assists verification and mapping, reasons over retrieved constraints, and drafts alerts or emails. It does not calculate inventory, run Monte Carlo simulation, solve optimization, or approve consequential actions.
- **Small and medium-sized enterprise (SME):** A business with fewer financial, analytical, and operational resources than a large enterprise, with thresholds varying by jurisdiction.
    - **Use and significance:** SMEs are a primary target because many rely on supplier emails, spreadsheets, search alerts, and periodic meetings. A narrow product-and-supplier pilot can provide earlier analysis without requiring a large risk-intelligence team.
- **Model Context Protocol (MCP):** An interoperability protocol through which an AI application can access defined resources, prompts, and tools through structured interfaces.
    - **Use and significance: In the current prototype, MCP may support read-only queries or draft-email creation; emails remain reviewable drafts. A future MCP connector may send the exact approved draft only after human review and approval. Authentication, authorization, validation, least privilege, and approval remain the application's responsibility.**
- **Natural language processing (NLP):** Methods that allow computers to analyze, structure, and generate human language.
    - **Use and significance:** NLP supports language detection, relevance classification, entity and relation extraction, summaries, alerts, and context-aware supplier or customer email drafts. It turns fragmented text into structured, reviewable information.

3. Detailed End-to-End Execution Workflow

The following sequence expands Section II.D and Section III of the white paper. Every transition records provenance, confidence, timestamp, user or agent identity, and version where applicable.

Step 1 - Establish authority and scope

A company administrator identifies data owners and decision approvers, then defines the permitted systems, business units, products, suppliers, locations, fields, users, retention periods, and approval thresholds. Connections operate with least-privilege access. The system records purpose, consent, access policy, and data lineage.

Step 2 - Upload or synchronize authorized company data

The company supplies relevant products, parts, bills of materials, supplier sites, qualified alternatives, inventory, in-transit stock, purchase and customer orders, production schedules, capacities, yields, logistics routes, lead times, costs, contracts, and service priorities. A pilot may begin with secure CSV templates; later versions may use read-only enterprise integrations.

Step 3 - Validate and normalize the data

The ingestion layer standardizes identifiers, units, currencies, dates, and time zones and detects duplicates, missing links, stale balances, impossible values, and contradictory supplier records. Significant quality problems are quarantined or presented to an authorized user. Unverified records do not silently influence high-impact recommendations.

Step 4 - Build the operational baseline

Validated records form a company-specific knowledge graph and a versioned digital-twin snapshot. Users inspect supplier-to-order paths, confirm alternatives and routes, check data freshness, and preserve the precise state used for later calculations.

Step 5 - Detect and verify an external disruption

The non-AI Scout agent polls approved RSS news feeds, de-duplicates entries, and stores source metadata; it does not judge relevance or verify claims. An LLM-based agent then filters company relevance and extracts structured event fields. A Verifier compares independent sources, checks temporal and geographic consistency, identifies contradictions, and flags unsupported or low-confidence claims for review.

Step 6 - Map exposure and calculate the no-action impact

The verified event is resolved against internal suppliers, sites, routes, ownership, parts, products, facilities, and customer commitments. Graph traversal identifies affected paths. Deterministic services and probabilistic simulation estimate inventory coverage, shortage and delay windows, production interruption scenarios, service exposure, incremental cost, and uncertainty if the company takes no action.

Step 7 - Generate and compare mitigation strategies

RAG retrieves current approved suppliers, qualified substitutes, contracts, policies, capacities, routes, and previous responses. The LLM-based Planner proposes candidate actions. A deterministic feasibility checker rejects options that violate qualification, capacity, commercial, legal, export-control, sanctions, budget, cybersecurity, or approval constraints. Simulation and optimization compare the remaining options.

Step 8 - Review, revise, and approve

A named decision-maker reviews sources, affected paths, enterprise records, freshness, assumptions, uncertainty, calculations, and alternatives. The user may correct data, change assumptions, edit an action, select another option, or rerun the scenario. Approval records the exact scope, budget, recipients, message version, and decision owner.

Step 9 - Execute controlled actions

The current implementation releases only internal, policy-permitted outputs after approval, such as creating procurement or planning tasks, notifying an internal team, or saving an approved supplier-status email as a reviewable draft. Chainsilience AI does not send external emails. In a future version, an MCP connector may transmit the exact approved draft only after an authorized reviewer approves its content, recipients, and scope. Every action, owner, deadline, response, confirmation, failure, and status change is recorded.

Step 10 - Observe outcomes and learn

The dashboard distinguishes the no-action baseline, the projected mitigated outcome, and the realized outcome. Supplier confirmations, inventory changes, shipment events, production records, receipts, and order data update the twin. Projected benefit is not reported as realized improvement until operational evidence supports it.

4. DSAI Components and Responsibility Boundaries

Chainsilience AI uses a hybrid DSAI architecture. The following map prevents the LLM from being credited with calculations or authority that belong to other components.

Component	Primary use in Chainsilience AI	Boundary / significance
Non-AI Scout and connectors	Poll approved RSS news feeds and retrieve minimum required internal fields.	The Scout does not classify, reason, verify, or generate; LLMs do not scrape or authenticate data sources.

Component	Primary use in Chainsilience AI	Boundary / significance
NLP and LLM agents	Filter relevance, extract event facts, compare claims, assist entity mapping, propose options, explain results, and draft communications.	Outputs remain probabilistic and evidence-linked; consequential actions require controls and approval.
RAG	Retrieve verified evidence and authorized company records before summaries, explanations, or mitigation proposals are generated.	Reduces unsupported output and makes recommendations traceable to current records.
Knowledge graph	Store multi-tier dependency relationships, provenance, confidence, and temporal validity.	Supports explainable company-specific exposure paths rather than flat keyword alerts.
Digital twin	Represent the operational state at a known time and provide reproducible scenario snapshots.	Separates current state, scenario assumptions, and later realized observations.
Deterministic calculations	Calculate days of cover, shortage dates, lead-time effects, costs, and constraint checks.	Numbers are recalculated outside the LLM.
Probabilistic simulation	Estimate scenario frequency and uncertainty through Monte Carlo sampling.	The Analyst agent supplies validated parameters and explains results; it does not perform the numerical simulation.
Optimization service	Compare feasible actions across service, cost, residual risk, and implementation time.	Only qualified and policy-compliant candidates enter the solver.
Policy gateway and approval	Enforce permissions, action limits, recipients, budgets, and human authorization.	MCP or agent orchestration is not itself a security boundary.

4.1 Component and AI-agent separation of duties

Component / agent	Permitted responsibility	Prohibited by default
Scout (non-AI)	Poll approved RSS news feeds; de-duplicate and store candidate items with source metadata.	Infer relevance, verify claims, or access sources outside the approved RSS allowlist.
Extractor	Produce a structured schema with cited evidence spans.	Fill missing fields with unsupported content.
Verifier	Compare sources and flag unreliable, contradictory, unsupported, or low-confidence claims.	Guarantee that a report is true or false without adequate corroboration.
Mapper	Resolve event entities against the authorized company graph.	Create high-impact graph facts from an unresolved match.
Analyst	Request deterministic and probabilistic scenarios and interpret returned results.	Override calculators or simulate numerically inside the LLM.
Planner	Propose evidence-grounded candidate mitigations.	Rank or select candidates using quantitative impact metrics, commit purchases, change suppliers, or bypass feasibility checks.
Communicator	Draft internal alerts and supplier or customer messages.	Send or dispatch external emails; outputs remain reviewable drafts.

4.2 RAG-grounded mitigation planning

During mitigation planning, RAG retrieves only authorized and current records relevant to the verified disruption. These records may include approved suppliers, qualified substitute components, contracts, company policies, available capacities, logistics routes, customer priorities, and previous disruption responses. The LLM-based Planner reasons over this retrieved context to propose candidate actions. Deterministic feasibility checks, simulation, and optimization then validate constraints and compare service recovery, cost, implementation time, and residual risk. Missing, stale, or conflicting records trigger review rather than an unsupported recommendation.

5. Data, Graph, and Digital-Twin Specification

5.1 Input data boundary

Layer	Representative fields	Primary control
External evidence	Articles, government advisories, earthquake and weather feeds, trade notices, port and carrier updates, company disclosures, licensed datasets.	Source allowlist, licensing, hashing, timestamping, language detection, deduplication, provenance.
Enterprise master data	Suppliers, sites, products, parts, materials, bills of materials, ownership, approved alternatives.	Role- and field-level access; validation; change history.
Operational state	Inventory, in-transit stock, orders, schedules, capacities, yields, routes, lead times, costs, priorities.	Freshness checks, snapshot versioning, purpose limitation.
Reasoning artifacts	Events, entity matches, graph paths, assumptions, scenario inputs, results, recommendations.	Evidence links, confidence decomposition, schema validation, version control.
Action records	Approvals, drafts, recipients, tasks, tool results, confirmations, failures.	Policy enforcement, immutability, audit, retention.

5.2 Event extraction schema

A relevant external item is converted into a structured event object containing: event type; source and evidence span; affected organization, facility, location, route, commodity, or product; start and observation time; expected duration or recovery range; severity indicators; reported quantities; source reliability; extraction confidence; contradiction status; and review state. Unknown values remain unknown rather than being completed by the LLM.

5.3 Knowledge-graph model

Graph element	Examples	Why it matters
Entities	Supplier, Site, Part, Material, Product, Route, Port, Carrier, Inventory Node, Production Line, Order, Customer, Event, Source.	Provides the objects needed to connect a disruption to operations.
Relationships	SUPPLIES, PRODUCED_AT, REQUIRES, SHIPS_VIA, STORED_AT, FULFILLS, LOCATED_IN, OWNED_BY, AFFECTS.	Represents multi-tier dependency paths that a flat database alert may hide.
Evidence and confidence	Source record, cited passage, extraction confidence, entity-match confidence, reviewer confirmation.	Allows every externally inferred relationship to be challenged and traced.
Temporal validity	Effective-from, effective-to, observation time, ingestion time.	Distinguishes a current supplier or route from an expired relationship.

Example exposure path

Earthquake -> Taiwan industrial area -> Supplier A site -> controller chip -> three products -> seven customer orders. The path explains why a geographically severe event is operationally relevant to this specific company.

5.4 Minimum digital-twin state

Entity or state	Minimum scenario fields
Supplier and site	Capacity, recovery curve, qualification status, concentration, ownership.
Part and material	BOM usage, compatibility, substitute status, yield loss, qualification time.
Inventory	On-hand, committed, quarantined, in-transit, expiry, allocation.
Route	Mode, lane, transit time, cost, schedule, chokepoints, customs constraints.
Production	Line, schedule, minimum run, yield, tooling, labor, alternate capacity.
Order and customer	Quantity, due date, priority, service commitment, penalty.

Twin updates are event-sourced. Each change records its source, effective time, ingestion time, previous value, and authorizing identity. Scenario runs use frozen versions so users can reproduce why a recommendation was made.

6. Quantitative Risk and Mitigation Logic

6.1 Risk decomposition

The normalized triage score ranks attention; it is not a probability. For event e and affected item i, the white paper combines evidence confidence (C), hazard severity and duration (H), exposure (X), vulnerability (V), time criticality (T), and mitigation strength (M) using versioned weights. Inputs are scaled to [0, 1], and the weighted result is clipped to [0, 1].

Interpretation rule

Risk rank answers 'what should be reviewed first?' Production-stoppage probability answers 'how often does a protected line fail under the stated scenario distributions?' These outputs use different metrics and must not be conflated.

6.2 Operational calculations

Measure	Definition	Decision use
Days of cover (DOC)	Available qualified inventory divided by protected daily demand, adjusted for quarantine, yield, expiry, and allocation.	Shows how long operations can continue before replenishment is required.
Projected shortage date	First period in which cumulative qualified supply is lower than cumulative protected demand.	Identifies the decision deadline and affected orders.
Expected incremental cost (EIC)	Premium purchase + logistics + qualification + changeover + penalties - avoided loss.	Compares mitigation expense with expected disruption loss.
Stoppage scenario frequency	Fraction of Monte Carlo runs in which a protected line cannot meet its minimum schedule.	Quantifies uncertainty under explicit duration, transit, yield, demand, and alternative-capacity distributions.
Action utility	Service benefit - cost - residual risk - implementation time, using visible policy weights.	Supports service-first, cost-first, or balanced comparison without hiding trade-offs.

6.3 Candidate action classes and controls

Action class	Required feasibility checks	Approval owner
Expedite or reroute	Lane, carrier, customs, capacity, cost, schedule.	Logistics owner
Reallocate inventory	Customer priority, contract, protected demand, service impact.	Operations owner
Switch or split supplier	Technical qualification, capacity, legal, sanctions, export control, IP, cybersecurity.	Cross-functional authority
Reschedule production	Line, tooling, labor, yield, due dates, changeover.	Plant planner
Customer or supplier notice	Verified facts, disclosure limits, commitments, recipients, approved wording.	Account or procurement owner

6.4 Decision-state reporting

Displayed state	Meaning
No-action baseline	Estimated impact if no mitigation is implemented under the selected snapshot and assumptions.
Projected mitigated outcome	Expected result of the approved strategy before sufficient operational evidence has arrived.
Realized outcome	Observed result supported by confirmations, inventory, shipment, production, receipt, and order records.

7. Explainability, Security, and Human Oversight

7.1 Evidence chain presented to the user

Interface card	Required content
Event facts	What happened, where, when, reported duration, and unresolved fields.
Evidence and source quality	Retrieved sources, cited passages, contradictions, and reliability indicators.
Affected graph path	The supplier, site, route, part, product, production, and order links that create exposure.
Operational assumptions	Data freshness, scenario inputs, distributions, user edits, and model versions.
Quantitative impact	Inventory coverage, shortage dates, delays, cost ranges, scenario frequency, and uncertainty.
Response options	Benefits, costs, implementation time, prerequisites, reversibility, owner, residual risk, and approval state.

7.2 Security and confidentiality controls

Sensitive company data - including suppliers, routes, products, orders, inventory, production schedules, prices, forecasts, and customer commitments - remains within the protected company environment and is used only for authorized supply-chain analysis. Controls include encryption, tenant isolation, role- and attribute-based access, secret management, source allowlists, data minimization, prompt and input sanitization, network egress restrictions, immutable audit events, retention schedules, backups, incident response, and periodic security testing. Internal data must not be disclosed to third parties or used to train external AI models without explicit company consent.

7.3 Approval and audit requirements

In the current implementation, external emails remain editable drafts even after a reviewer approves their content; Chainsilience AI does not send them. Procurement commitments, supplier changes, and customer promises require approval by named roles. A future MCP sending capability may transmit only the exact approved draft after explicit review of its recipients, content, and scope. The approval record includes the reviewer, time, exact action or message version, recipients, budget, evidence, and downstream result. Expired, changed, or previously consumed approvals are rejected.

7.4 Failure and escalation behavior

- **Low-confidence event:** Request verification instead of generating a definitive high-severity warning.
- **Unresolved entity:** Place the match in a review queue; do not silently create a graph relationship.
- **Stale internal data:** Display the last refresh time and block or qualify calculations that depend on freshness.
- **Conflicting sources:** Show the conflict, retain both evidence paths, and request confirmation.
- **No feasible mitigation:** Escalate the constraint conflict instead of fabricating an alternative.
- **Tool or workflow failure:** Record the failure, preserve idempotency, notify the responsible owner, and avoid partial unauthorized action.

8. Pilot Evaluation and Implementation Plan

8.1 Meaning of the pilot

The pilot is a small, controlled validation project before company-wide deployment. It may cover one company or business unit, one product family, a limited supplier and route set, selected disruption categories, a fixed historical replay period, and a small group of operational users. The targets below are acceptance criteria, not achieved results.

8.2 Evaluation protocol

- Replay dated historical disruption events against frozen enterprise snapshots.
- Withhold articles published after each decision cutoff to prevent future-information leakage.

- Have two analysts independently label event fields and affected graph paths; adjudicate disagreements.
- Report synthetic rare-event scenarios separately from historical-event performance.
- Compare against keyword alerts plus manual analysis, LLM extraction without verification, and risk ranking without a digital twin.
- Use component-removal tests to measure the contribution of provenance, graph context, and deterministic simulation.
- Have analysts blind-score relevance, traceability, feasibility, and clarity; have operational users complete timed scenario tasks and a structured usefulness survey.

8.3 Proposed pilot metrics

Dimension	Metric	Pilot acceptance target
Detection	Recall within 24 hours; false alerts per day	>=85%; <=2
Extraction	Entity and relation F1	>=85%
Verification	Unsupported-claim rate	<2%
Mapping	Correct affected-path rate	>=80%
Risk ranking	Precision@K; rank correlation	Improve baseline
Production-stoppage probability	Brier score; calibration error	Improve baseline
Impact	Mean absolute error for delay and DOC	Threshold set by use case
Action	Feasible recommendation rate	>=90%
Workflow	Median alert-to-decision time	50% lower than baseline
Safety	Unauthorized action rate	0

Recall within 24 hours measures detection timeliness; F1 balances missed and incorrect extractions; Precision@K measures whether top-ranked risks are important; Brier score and calibration error assess probability quality; mean absolute error measures average prediction error; and DOC means days of cover.

8.4 Proposed 12- to 16-week pilot

Period	Primary work	Exit evidence
Weeks 1-3	Define ontology, data-sharing boundary, users, approval rules, baseline, and acceptance criteria.	Authorized scope, data dictionary, evaluation plan.
Weeks 4-7	Integrate a narrow supplier and product scope; build collection, relevance filtering, extraction, and verification.	Versioned events with cited evidence and quality checks.
Weeks 8-10	Add graph mapping and deterministic inventory and lead-time calculations.	Traceable affected paths and reproducible impact calculations.
Weeks 11-13	Add scenario comparison, RAG-grounded planning, dashboard, and reviewable communications.	Feasible options, approval flow, audit records.
Weeks 14-16	Run replay tests, red-team exercises, user acceptance, and metric review.	Pilot report with achieved results separated from targets.

9. Illustrative Taiwan-Earthquake Scenario

This scenario expands the white paper's retrospective example. The 3 April 2024 Taiwan earthquake is a real evidence-based trigger, while Orion Controls, its dependencies, probabilities, quantities, costs, and allocation choices are fictional design-test inputs.

9.1 Scenario assumptions

Input	Illustrative value
Protected part	Controller chip used by three fictional product SKUs.
Primary dependency	Supplier A in Taiwan provides 42% of qualified wafer volume.
Inventory	12 days of usable coverage at protected demand of 8,000 units per day.
Alternative	Supplier B is approved for 25% incremental volume after a five-day changeover.
Mitigation cost	USD 180,000, with an illustrative range of USD 150,000-230,000.
Scenario output	71% of simulation runs show production stoppage within 21 days under the stated distributions.

9.2 Event-to-decision trace

Stage	System behavior	Evidence retained
Collect	The non-AI Scout polls approved RSS feeds for permitted earthquake and company-impact reports.	Source URL, hash, publication and ingestion times.
Filter and verify	Classify company relevance; extract location, time, severity, and affected entities; cross-check sources.	Cited passages, contradiction state, confidence components.
Map	Resolve the event to Supplier A's site and traverse chip, product, and order dependencies.	Entity matches and Supplier -> Part -> Product -> Order path.
Simulate no action	Estimate shortage day, recovery range, affected orders, and stoppage scenario frequency.	Frozen twin version, distributions, calculation and simulation versions.
Plan	Retrieve approved alternatives and propose wait, expedite, reallocate, and switch-25% options.	Supplier qualification, capacity, route, contract, and policy records.
Compare	Test constraints and compare service recovery, cost, time, reversibility, and residual risk.	Feasibility results, solver inputs, ranked alternatives.
Approve and communicate	A decision-maker selects the exact option and approves a reviewable supplier-status draft.	Reviewer, action scope, recipients, message version, timestamp.
Observe	Update the twin with supplier replies, shipment status, inventory, production, and order outcomes.	Projected versus realized results and before-after audit trail.

9.3 Example communication boundary

Reviewable supplier-status request

The Communicator may draft questions about facility status, available capacity, recovery process, allocation policy, revised lead times, and the next update time. It must not state that a three-week delay is confirmed unless retrieved evidence supports that claim. The procurement owner may edit and approve the draft, but Chainsilience AI does not send it. A future MCP connector may send only the exact approved version after explicit review; any resulting reply can then be retained as new evidence.

10. Limitations and Annex Control Notes

Limitation	Required interpretation or control
Undisclosed multi-tier dependencies	The system cannot map relationships absent from public evidence and authorized enterprise data; unresolved exposure remains visible as uncertainty.
Source truth	Cross-source verification can reduce unsupported claims but cannot guarantee that every source is correct.
Stale enterprise records	Digital twins decay; freshness and version must accompany every scenario.
Unprecedented events	Historical and synthetic evaluation may not represent novel combinations of shocks.
Qualification constraints	A publicly mentioned supplier or component is not operationally usable until enterprise, technical, legal, commercial, and cybersecurity checks pass.
Model and simulation uncertainty	LLM outputs are probabilistic; simulations inherit uncertain distributions; results require ranges, sensitivity analysis, and human judgment.
System authority	Chainsilience AI is decision support, not autonomous procurement, production, or external-communication authority.

Reference note: bracketed sources, standards, and literature supporting these concepts are listed in the bibliography of the revised Chainsilience AI white paper. This annex intentionally avoids adding claims beyond that paper.